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From Agentic Workflows to Frontier Questions

Previously, we examined why foundation models became useful (Topic 15), how retrieval and tools extend them (Topic 16), and what changes when those components become persistent workflows (Topic 17).

In this chapter, we ask a different question:

- if current LLM systems are powerful but incomplete,
- what kinds of ideas are most likely to matter next,
- and what should engineers believe without drifting into mythology?

Why This Chapter Matters

The future-AI conversation is often distorted by false choices:

- either LLMs already contain everything needed,
- or one replacement paradigm will soon sweep them away,
- or engineering judgment will dissolve into prompting.

This chapter takes a stricter view

The most credible future is not one universal architecture. It is a layered stack in which different mechanisms handle interface, explicit structure, prediction, physical action, and oversight.

Learning Objectives

After this lecture, students should be able to:

- explain why today's AI is best understood as a system stack rather than a standalone model,
- distinguish among foundation models, neuro-symbolic methods, world models, and embodied intelligence,
- describe why world models matter most when prediction and control are central,
- explain why embodiment matters for physical agency but not equally for every useful AI system,
- evaluate AGI and ASI claims using engineering standards of evidence, and
- explain why engineers remain responsible for constraints, verification, and accountability.

The Current Paradigm: Foundation-Model Pragmatism

The foundation-model era has already changed coding, search, content generation, and parts of engineering workflow. In practice, useful systems are no longer just "a model." They are models connected to:

- retrieval,
- memory,
- tools,
- environments, and
- human supervision.

Source: Bommasani et al. (2021): Foundation Models

Why System Design Matters So Much

Current LLM systems are strong at:

- language interface,
- pattern extraction,
- summarization,
- procedural decomposition.

They are weaker at:

- robust grounding,
- stable physical state tracking,
- reasoning about open-ended physical consequence.

Source: Wang et al. (2023): NEWTON; Qiu et al. (2025): PHYBench

The Future AI Stack as an Engineering Stack

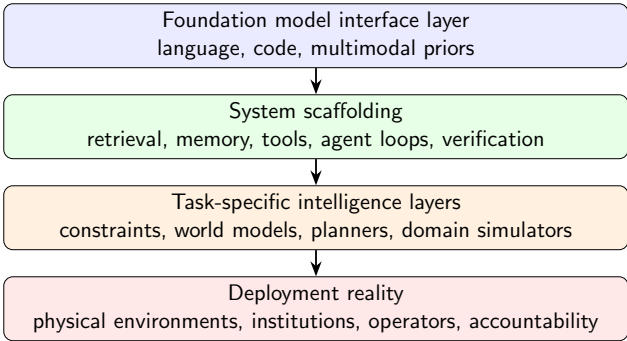


Figure: Recreated teaching figure: a layered view of likely future AI system design.

Interpreting the Stack

Main claim

The credible future is not “LLMs disappear” or “LLMs become everything.” Foundation models remain useful interface layers while other mechanisms become more important around them.

From an engineering perspective, the system boundary now includes:

- prompt structure,
- memory and retrieval,
- tool interfaces,
- verification layers,
- operators and institutions.

Why Language-First Systems Are Still Incomplete

A fluent model can imitate reasoning traces and summarize rules, but that is not the same as:

- keeping a stable world state,
- predicting physical consequences under intervention,
- recovering from novel physical failure,
- acting safely in real environments.

Engineering consequence

If the task involves explicit rules, physical prediction, or high consequence, language fluency alone is rarely enough.

What Neural-Symbolic AI Means

Definition

Neuro-symbolic AI combines learned subsymbolic representations with explicit symbolic machinery: rules, variables, programs, constraints, logic, or probabilistic reasoning.

Motivation:

- neural systems scale well in noisy, high-dimensional perception,
- symbolic systems preserve explicit structure and traceable inference,
- each side is strong where the other is weak.

Source: Besold et al. (2017); Colelough & Regli (2025)

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Why Explicit Structure Keeps Returning

Engineers care about domains where the structure is not optional:

- laws,
- design constraints,
- operating procedures,
- ontologies,
- program semantics,
- scientific priors.

In such settings, symbolic structure improves:

- auditability,
- checkability,
- data efficiency,
- controllability.

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Representative Neuro-Symbolic Systems

Representative systems make the idea concrete:

- **DeepProbLog**: learned perception inside probabilistic logic,
- **Neural Logic Machines**: relational and logical inductive bias,
- **NS-CL**: object-centric perception + semantic parsing + executable programs.

Source: Manhaeve et al. (DeepProbLog); Dong et al. (NLM); Mao et al. (NS-CL)

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The Modern Extension: Agentic AI as System-Level Hybridization

Topic 17 described modern agentic systems in terms of:

- skills,
- tool interfaces,
- orchestration logic,
- explicit constraints,
- human review checkpoints.

Careful claim

Many agentic systems are neuro-symbolic at the *system level*, even when their neural core is still a large language model rather than a classical theorem prover.

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Why Skills and Constraints Function Symbolically

When an engineer defines a reusable workflow, the engineer is not merely “prompting” a model.

The engineer is imposing symbolic structure on the task:

- what steps exist,
- which tools may be called,
- what counts as acceptable evidence,
- when the system must stop,
- when a human must intervene.

If those rules can be inspected and revised by humans, they are functioning like a symbolic layer around a neural substrate.

Why Natural Language Can Still Be Symbolic

Natural language carries symbolic structure when it is used to define:

- roles,
- branches,
- pass/fail criteria,
- escalation rules,
- scenario-specific constraints.

Important caution

The careful claim is not that every agentic system is a classical symbolic reasoner. The claim is that many useful agentic systems are neuro-symbolic in practice because neural models operate inside human-defined symbolic workflows.

Engineering Example: Design Code Checking

Where this matters in engineering

Imagine an AI assistant for design code checking. A purely neural system may summarize text well, but the workflow still needs explicit clauses, pass/fail constraints, traceable justification, and human review of edge cases.

That is exactly the kind of setting in which symbolic structure is not nostalgia. It is operationally useful.

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What Counts as a World Model?

The phrase **world model** is used in more than one way.

- In model-based reinforcement learning: a learned dynamics model that lets an agent imagine futures and plan in latent space.
- In the JEPA line: predictive latent representations that capture what is predictable and task-relevant without reconstructing every surface detail.

These communities overlap, but they should not be collapsed into a single slogan.

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The PlaNet and Dreamer Line

The model-based RL trajectory is already substantial:

- **PlaNet**: latent dynamics planning from pixels,
- **Dreamer**: behavior learned through latent imagination,
- **DreamerV3**: a much more scalable control program across many tasks.

This is not hype. It is a real research program with repeated empirical gains.

Source: Hafner et al. (PlaNet)

What Dreamer Adds Conceptually

Dreamer matters because it sharpens the idea of a world model:

- learn a latent state,
- imagine future trajectories in that latent space,
- optimize behavior using imagined rollouts rather than only expensive environment interaction.

Source: Hafner et al. (Dreamer)

LeCun's Critique of Pure Next-Token Prediction

LeCun's roadmap argues that next-token prediction alone may be an insufficient route to autonomous intelligence because intelligence also requires:

- hierarchical representation,
- prediction,
- planning,
- action over latent world state.

Source: LeCun (2022): A Path Towards Autonomous Machine Intelligence

I-JEPA and V-JEPA

I-JEPA and V-JEPA should be read as attempts to operationalize that roadmap:

- predict latent representations rather than reconstruct every pixel,
- emphasize what is predictable and semantically useful,
- move from image prediction to video and, increasingly, planning-relevant settings.

Source: Assran et al. (I-JEPA); Bardes et al. (V-JEPA)

Autoregression versus World Models

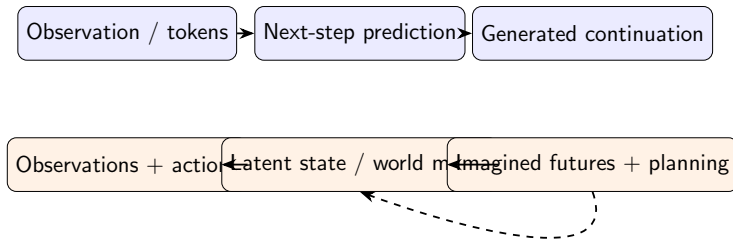


Figure: Recreated teaching figure: two predictive emphases.

How to Read the Figure

The top path asks:

- what comes next,
- how should the sequence continue,
- how do we maximize fluent continuation?

The bottom path asks:

- what state of the world am I in,
- what happens if I act,
- which imagined future should I prefer?

What World Models Buy You

World models are one of the strongest candidates for progress beyond language-only systems because they target:

- prediction,
- abstraction,
- planning,
- action.

They are especially compelling where:

- control matters,
- consequences matter,
- interventions change future observations.

Why the Evidence Is Still Narrower Than the Hype

The strongest evidence for world models is concentrated in:

- reinforcement learning,
- robotics,
- video representation,
- planning-heavy tasks.

Do not overclaim

It is not yet well supported to say that JEPA-style methods have already superseded autoregressive modeling across all of AI.

Source: Destrade et al. (2026): Value-guided JEPA Planning

V-JEPA 2 and the Move Toward Robotics

V-JEPA 2 is especially notable because it pushes from large-scale video prediction toward robotic planning with a small amount of robot-interaction data.

That matters because it shows a plausible route from:

- predictive latent representation,
- to planning,
- to embodied deployment.

Source: Assran et al. (2025): V-JEPA 2

What Embodiment Actually Means

Embodiment does not simply mean “having a camera” or “being multimodal.”

Definition

Embodiment means intelligence coupled to a body that acts in an environment, changes future observations through action, and must manage the sensorimotor consequences of its own behavior.

Source: Brooks (1990): Elephants Don't Play Chess

Why Physical Deployment Changes the Problem

Some useful AI systems do not need bodies in the strong sense:

- theorem assistants,
- schedulers,
- CAD copilots,
- document-review systems.

But once the world can push back through friction, delay, occlusion, contact, or damage, text-only competence becomes a weak proxy for intelligence.

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Why Robotics Put Embodiment Back at the Center

Modern robotics research explains why embodiment returned to the center of AI discussion:

- RT-1 and RT-2 showed broad robot control with transformer-style architectures,
- Open X-Embodiment demonstrated positive transfer across robots and institutions,
- OpenVLA and Octo pushed the generalist robot policy idea further.

Source: Brohan et al. (RT-2); Open X-Embodiment Collaboration; OpenVLA

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From Language Interface to Physical Agency

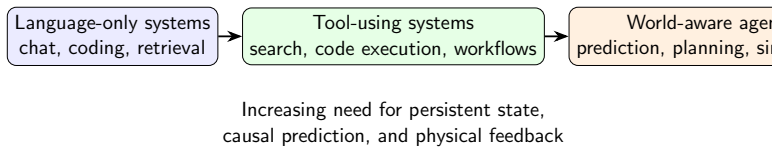


Figure: Recreated teaching figure: from interface intelligence to physical agency.

Biological Analogies: Use Carefully

The biological examples are useful pedagogically if interpreted carefully:

- paramecium and amoebae show adaptive environment-coupled behavior below brains,
- crows and dolphins show that sophisticated cognition is not confined to primates.

The lesson is not that robots are about to become animals.

The lesson is that intelligence is biologically plural and strongly shaped by bodies and niches.

What Embodiment Does *Not* Imply

Do not universalize from robotics

Embodiment is necessary for robust physical agency, but it is not obviously necessary for every economically useful form of AI competence.

The stronger claim is domain-specific:

- when the world can push back through physics,
- embodiment becomes increasingly important,
- but not every useful AI system must become embodied.

Engineering Takeaway on Embodiment

If the task is:

- closed-loop,
- safety-critical,
- contact-rich,
- irreversible,

then embodiment becomes a first-order engineering question rather than a philosophical embellishment.

Examples

maintenance robotics, navigation, manipulation, human-robot interaction, industrial autonomy.

Why AGI and ASI Are Slippery Terms

Artificial general intelligence and artificial superintelligence are among the least disciplined terms in public AI discourse.

Problems:

- no consensus operational benchmark,
- performance can be confused with intelligence,
- impressive task demonstrations can be over-generalized into species-level narratives.

Source: Morris et al. (Levels of AGI); Chollet (2019)

Why Capability Talk Jumps Too Fast

The literature discussed here does *not* demonstrate a system with all of the following together:

- broad autonomous competence across open-ended environments,
- stable self-improvement under weak oversight,
- robust goal alignment under superhuman capability,
- reliable control under distribution shift.

Those are separate requirements, and the evidence does not yet show them together.

What Serious Near-Term Work Actually Studies

The most serious work near this area is not proof of ASI. It is **oversight research** motivated by the possibility that future systems may outstrip direct human inspection.

Examples:

- Constitutional AI,
- weak-to-strong generalization,
- debate and consultancy protocols.

Source: Bai et al. (Constitutional AI); Burns et al. (Weak-to-Strong)

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Engineering Standard of Evidence

A practical rule

A strong future-AI claim should be evaluated the same way any engineering claim is evaluated: what was demonstrated, under what conditions, with what controls, with what failure modes, and with what verification mechanism?

This standard is deliberately stricter than hype-cycle rhetoric.

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What the Literature Does Not Yet Justify

The responsible teaching stance is neither ridicule nor prophecy.

It is to say clearly:

- the evidentiary burden is extreme,
- current results are interesting but partial,
- oversight and governance are more concrete than superintelligence storytelling.

The Likely Next Decade: Diversification of the Stack

From an engineering point of view, the likely next decade is not a clean handoff from LLMs to one replacement architecture.

It is a diversification of the stack:

- foundation models as interface layers,
- neuro-symbolic methods where explicit structure matters,
- world models where prediction and planning matter,
- embodiment where physical deployment matters.

Distinction Sheet I

Theme	Established	Strong hypothesis
Current LLM systems	Powerful for language, code, retrieval, and tool-using workflows	Better scaffolding can keep extending usefulness in many professional settings
Neuro-symbolic AI	Hybrid systems can improve structured reasoning and explicit constraint handling	Constraint-aware hybrids will become more important in engineering-grade AI
World models	Learned latent dynamics help control and planning in RL and robotics	Predictive state models are necessary for robust physical agency

Distinction Sheet II

Theme	Established	Speculative
Embodiment	Physical interaction changes the competence required	Every useful future AI system must be embodied
AGI / ASI	There is no consensus operational benchmark	Near-term ASI is already an established engineering forecast

The Less Credible Forecast

The less credible forecast is the one preferred by hype cycles:

- one architecture becomes universally intelligent,
- human oversight fades away,
- engineering collapses into prompting.

Why this is weak

The literature supports a more demanding picture in which future AI systems become more heterogeneous, more infrastructure-like, and more dependent on careful interface design, deployment boundaries, and verification.

Final Message

The frontier after current LLM systems is real.

- Neuro-symbolic AI says that intelligence without explicit structure is often not enough.
- World-model research says that prediction and planning over latent state may matter more than surface continuation when action matters.
- Embodiment says that intelligence tested only in language is not the same as intelligence tested in the world.
- AGI/ASI discourse says that capability talk without measurement and oversight is mostly theater.

Engineers Stay Above AI

Chapter conclusion

Regardless of paradigm, engineers stay above AI because engineers still define the task, the constraints, the acceptable evidence, the failure boundaries, and the accountability chain.

That is the design principle that should survive every architecture transition discussed in this course.